

KUZ' MOVKA, Russian Federation

WMO: 237880

Lat: **62.3145N**Lon: **92.1132E**Elev: **62**StdP: **100.58**Time Zone: **7.00 (E07)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-48.8	-45.7	-51.4	0.0	-48.8	-48.3	0.0	-45.6	6.2	-13.6	5.3	-12.7	0.1	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	29.2	18.8	27.1	17.9	24.9	17.3	20.3	26.4	19.2	24.8	18.1	23.0	1.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.2	22.4	17.2	12.3	21.4	16.1	11.5	20.6	59.0	26.7	55.0	24.9	51.5	22.8	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.8	5.0	4.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.7	-28.7	-23.0	-12.2	-3.3	3.9	13.2	17.2	13.2	6.4	-3.7	-17.5	-23.2
	DBStd	17.00	11.01	9.54	8.38	6.09	4.17	5.12	3.32	3.44	4.27	5.94	10.65	11.03
	HDD10.0	5848	1199	924	689	398	194	28	1	11	123	424	826	1030
	HDD18.3	8448	1457	1158	948	648	447	165	60	162	357	683	1076	1288
	CDD10.0	480	0	0	0	0	6	123	224	110	17	0	0	0
	CDD18.3	39	0	0	0	0	0	11	26	3	0	0	0	0
	CDH23.3	761	0	0	0	0	3	286	408	61	3	0	0	0
	CDH26.7	194	0	0	0	0	0	75	111	9	0	0	0	0
(14) Wind	WSAvg	1.4	0.9	1.0	1.3	1.9	1.8	1.5	1.3	1.3	1.5	1.7	1.2	0.9
(15) Precipitation	PrecAvg	749	50	41	44	47	57	66	66	78	72	88	76	64
	PrecMax	958	91	65	75	77	105	127	132	127	121	142	153	107
	PrecMin	519	5	18	16	18	24	16	12	25	11	24	38	27
	PrecStd	115	20	12	16	17	21	24	26	25	25	21	25	22
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.2	-1.5	5.6	12.6	22.4	31.0	31.8	27.8	22.1	8.2	1.4	-1.6
		MCWB	-4.9	-2.2	1.4	5.7	12.5	19.0	20.4	18.8	15.7	5.2	0.5	-2.4
	2%	DB	-6.8	-3.5	2.5	9.8	17.9	28.9	29.8	24.6	18.4	6.5	-0.6	-3.8
		MCWB	-7.4	-4.4	0.2	3.9	10.4	17.7	19.7	17.7	13.6	4.1	-1.5	-4.5
	5%	DB	-9.6	-6.5	0.8	7.0	14.1	27.0	27.9	22.8	15.1	4.6	-1.9	-5.5
		MCWB	-10.3	-7.2	-0.8	2.8	8.3	17.0	18.8	16.8	11.9	3.3	-2.7	-6.1
	10%	DB	-12.0	-9.2	-0.6	5.0	11.3	24.3	25.8	20.6	13.0	2.9	-3.3	-7.6
		MCWB	-12.5	-9.8	-2.0	1.4	6.2	16.2	17.5	15.7	10.5	1.8	-3.9	-8.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.8	-1.9	2.1	6.1	13.8	20.9	22.3	20.5	16.5	6.0	0.6	-2.2
		MCDB	-4.3	-1.5	4.3	11.6	20.7	27.1	28.7	24.9	21.3	7.3	1.2	-1.6
	2%	WB	-7.3	-4.2	0.4	4.4	11.1	19.5	21.0	18.8	14.2	4.4	-1.2	-4.4
		MCDB	-6.8	-3.4	2.1	8.4	16.4	26.0	27.8	23.2	16.9	5.9	-0.6	-3.8
	5%	WB	-10.3	-7.2	-0.7	3.2	9.0	18.2	19.8	17.7	12.5	3.1	-2.7	-6.1
		MCDB	-9.7	-6.5	0.8	6.2	13.7	24.4	25.7	21.5	14.9	4.5	-2.0	-5.6
	10%	WB	-12.5	-9.9	-1.9	2.1	6.7	16.8	18.7	16.4	10.7	1.9	-3.9	-8.3
		MCDB	-11.9	-9.3	-0.7	4.5	10.6	22.9	24.1	19.9	12.7	2.9	-3.2	-7.7
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	10.9	13.7	12.0	10.8	14.0	13.3	11.9	9.1	6.5	9.4	9.6
		MCDBR	9.3	9.2	11.0	14.4	16.8	19.5	18.1	16.5	13.3	7.2	8.7	7.1
		MCWBR	9.0	9.0	9.2	9.5	9.7	9.1	8.9	9.9	8.4	5.6	8.4	6.7
	5% WB	MCDBR	9.0	9.0	9.7	12.5	15.5	15.9	15.2	14.1	12.3	6.9	8.3	7.1
		MCWBR	8.8	8.8	8.2	8.6	9.6	8.0	8.3	9.1	8.1	5.6	8.1	6.8
(40) Clear-Sky Solar Irradiance	taub		0.256	0.297	0.331	0.356	0.372	0.387	0.432	0.411	0.351	0.307	0.241	0.229
	taud		1.901	1.970	1.982	2.016	2.163	2.382	2.246	2.268	2.398	2.193	1.951	1.865
	Ebn at Noon		356	611	744	814	839	836	779	760	748	615	375	199
	Edh at Noon		41	80	115	137	129	107	120	108	78	63	36	22
(44) All-Sky Solar Radiation	RadAvg		0.28	1.11	2.46	4.06	5.08	5.51	4.94	3.40	2.03	0.82	0.36	0.13
	RadStd		0.04	0.10	0.26	0.46	0.46	0.57	0.46	0.36	0.30	0.16	0.07	0.02

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	+2.96	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page